

Project ID :

25-26J-191

1. Topic (12 words max)

A Cloud-Integrated IoT and ML-Based Waste Management System for Railway Stations

2. Research group the project belongs to

AIMS - Autonomous Intelligent Machines and Systems

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

Railway stations are a central part of Sri Lanka's public transport system, connecting both remote and urban regions and facilitating the daily commute for thousands of its residents. Apart from their logistical role, railway stations are also an entry point for domestic and international tourists, particularly those using Sri Lanka's iconic scenic routes such as Colombo to Ella and Kandy to Badulla [1]. Consequently, the sanitary conditions of the areas significantly affect public health, commuter satisfaction, and the country's overall tourist image [2].

Despite such importance, waste disposal at Sri Lankan railway stations remains entirely manual, reliant on pre-determined cleaning schedules and manpower, irrespective of real-time pedestrian volume or rubbish accumulation. Platforms are likely to accumulate random litter like plastic containers, paper, food packaging, and beverage bottles especially during rush hours and holidays. Things are worse in open-air platforms, as wind carries light litter and heavy use exceeds static cleaning [3]. In a tropical climate, such uncontrolled litter can quickly lead to foul odor, insect infestation, overflowing bins and microbial risk, and substantially worsen hygiene and passenger experience [4].

Besides, intelligent waste monitoring or robot-based cleaning systems have not yet been

implemented in these environments. There are no sensor systems for garbage detection or localization, no camera systems for material type classification, and no autonomous machinery to handle dynamic cleaning demands. Such a lack of technological utilization indicates a major shortfall in adapting to smart city ideals and public service reform [5], [6].

In addition to influencing commuter convenience, poor sanitation also negatively influences tourists' perception of the country, something of specific interest considering Sri Lanka's stance as a cultural and eco-friendly tourist destination. Therefore, the absence of smart, adaptive, and automatic waste management facilities in railway stations is a complex, open-ended, and urgent social problem that intersects with public health, sustainability, tourism, and tech policy.

References

- [1] P. K. Ratnayake and G. K. A. D. S. Gunaratne, "Railway transport as a tourism product in Sri Lanka," *Sri Lanka Journal of Social Sciences*, vol. 41, no. 2, pp. 97–107, 2018.
- [2] M. de Albuquerque Meneguzzi, L. S. de Almeida, and A. T. da Silva, "The impact of sanitation infrastructure on tourism satisfaction: A systematic review," *Journal of Tourism Analysis*, vol. 29, no. 2, pp. 133–148, 2021.
- [3] C. Gunasekara and R. Silva, "Analysis of current municipal solid waste management systems in Sri Lanka: Problems and perspectives," *International Journal of Environmental Research*, vol. 15, no. 3, pp. 112–121, 2021.
- [4] H. M. N. Bandara et al., "Municipal solid waste management in Sri Lanka: Current practices and challenges," *Asian Journal of Environment & Ecology*, vol. 13, no. 4, pp. 1–14, 2021.
- [5] R. Ahmed, M. S. Islam, and A. Hoque, "Smart city waste management systems: A review of technologies and challenges," *IEEE Access*, vol. 10, pp. 56512–56525, 2022.
- [6] A. Hannan, A. M. Hoque, and M. S. Hossain, "Design and implementation of an IoT-based smart waste bin for real-time monitoring and data analytics," *Procedia Computer Science*, vol. 194, pp. 333–342, 2021.

6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

Our solution provides an intelligent, automated waste management system designed for railway stations in Sri Lanka. The system integrates robotics, IoT, and machine learning technologies to automate waste collection, classification and segregation with minimal human involvement.

The system starts with an autonomous garbage collecting robot that detects and picks up waste items scattered around the station using a camera and object detection model. This robot is equipped with an internal waste bin that stores the collected garbage. To ensure continuous operation, the robot features an internal bin monitoring system that tracks the fill level using ultrasonic sensors.

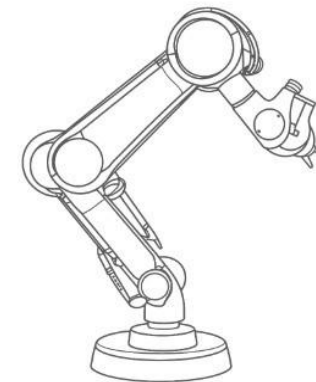
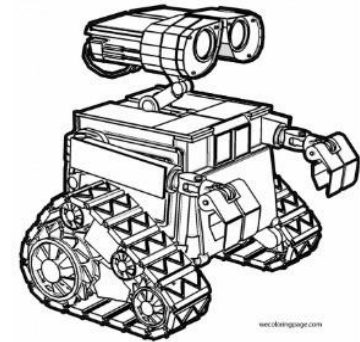
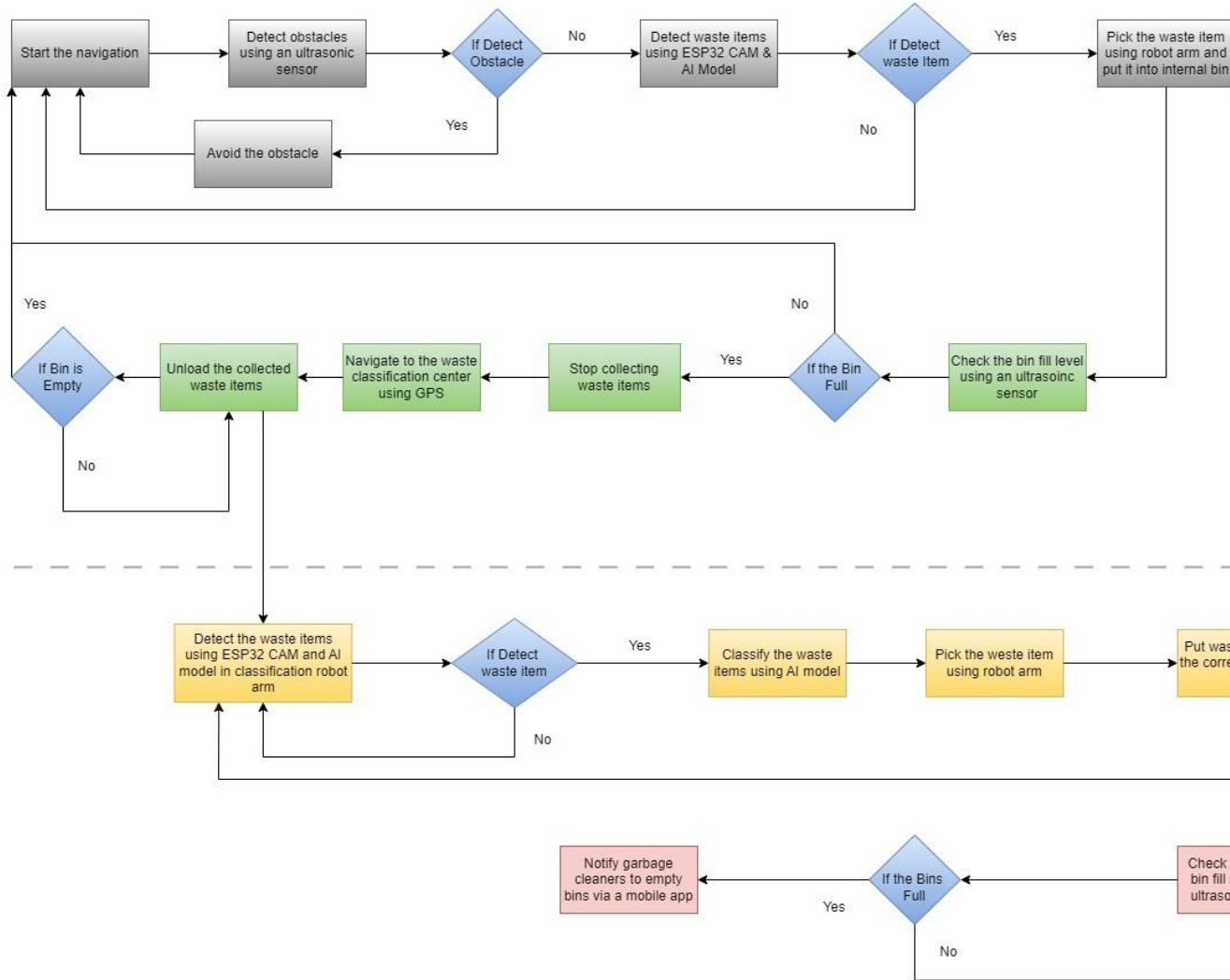
When the bin reaches full capacity, the robot autonomously navigates to a designated disposal station. At the disposal point, the robot empties the collected waste onto a classification platform. Here a custom trained machine learning model is used to classify the waste into four categories: Plastic, Paper, Glass, and Metal. The model is trained using a diverse dataset of railway station waste images to ensure high accuracy.

Once classified, a robotic classification arm picks each item and sorts it into the appropriate disposal bin based on the category. This ensures proper waste segregation and supports efficient recycling practices.

Overall, our solution includes four main components:

1. Mobile Robot Navigation System
2. Garbage Collecting Arm
3. Garbage Classification Model
4. Classification Robotic Arm

This fully integrated system promotes cleaner public spaces, reduces manual labor and supports Sri Lanka's smart city and sustainability goals.



7. Brief description of specialized domain expertise, knowledge, and data requirements
(300 words max)

To successfully develop this integrated autonomous garbage collection and classification system, expertise is required across several specialized domains. In Embedded Machine Learning (Cloud AI), requires skills in training lightweight machine learning models that can be run on limited devices such as the ESP32-CAM. Programmers must be familiar with Yolo, quantization, and optimization procedures for embedded inference. Familiarity with developing custom datasets, namely annotated photos for different kinds of waste typical in Sri Lanka is important in developing efficient detection and classification models.

In the field of Robotics and Mechatronics, proper manipulation of the robotic arm for waste collection as well as waste assortment requires a specialization in servo motor manipulation, mechanical engineering, and a grasp of kinematics with the ability to integrate sensors and actuators in an effective feedback loop for real-time operation. Also, in building a reliable communication system using ESP32 and ultrasonic sensors requires IoT development knowledge. The system must handle real-time data from sensors (e.g., garbage bin fill levels) and push updates to cloud services with minimal latency.

Mobility for the system is dependent on Autonomous Navigation and Obstacle Avoidance experience, including path-planning algorithms, ultrasonic sensor interpretation, and making navigation decisions in real-time without preprogrammed routes. Finally Mobile Application Development & Cloud Integration knowledge is required for developing mobile applications (Flutter) and cloud infrastructure needs to be present for a responsive dashboard for tracking bins and notification to be attained.

Data needs for the project also include high-quality annotated waste category photograph datasets [1], simulation-type datasets [2] modeling real-world conditions of operation like streets, railway stations, as well as exact sensor calibration data [3][4][5] for effective bin level detection with navigation reliability.

Sample Datasets:

[1] [Open-source computer vision datasets](#)

[2] [Waste classification data](#)

[3] [Trash detection object detection dataset](#)

[4] [Garbage image dataset](#)

[5] [Construction and demolition waste objects](#)

8. Objectives and Novelty

Main Objective			
This research aims to design and develop a cloud-integrated IoT-based autonomous robotic system capable of detecting, collecting, and classifying waste in real-world railway station environments, with the aim of improving sanitation efficiency, minimizing manual intervention, and aligning with smart city and sustainability goals.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
Perera W.B.N. IT22566348	Implement a robotic arm system capable of detecting and collecting visible garbage from the environment and depositing it into the internal bin of the robot using onboard object detection.	<u>Autonomous Garbage Detection and Collection</u> 1) Real-Time Garbage Detection: Use a camera module paired with a custom-trained object detection model to identify garbage items in the robot's path through live video processing. 2) Robotic Arm Activation: Trigger the robotic arm mechanism automatically when garbage is detected within reachable range. 3) Deposit Waste into Internal Bin: Place the collected waste securely into the onboard garbage bin, readying the robot for the next collection task.	Unlike traditional robots that follow fixed routes or need manual input, this component enables full autonomy by detecting garbage in real time and automatically activating the robotic arm to collect it without any human involvement.

Sendanayaka H.K. IT22582324	Enable the mobile robot to navigate autonomously with obstacle avoidance, find the disposal area using location awareness after monitoring its internal bin level and trigger autonomous unloading .	<u>Autonomous Navigation and Unloading Process</u> 1) Autonomous Navigation with Obstacle Avoidance: Implement intelligent movement logic allowing the robot to navigate complex station environments and avoid obstacles using real-time sensor input. 2) Internal Bin Monitoring and Fullness Detection: Continuously monitor the internal bin's fill level using distance sensors and determine when it reaches the unloading threshold. 3) Location-Based Unloading and Task Reset: Identify the disposal area using GPS or fixed-location mapping, trigger the unloading process upon arrival, and resume garbage collection afterward.	Unlike conventional systems that rely on manual control or fixed schedules, this component enables a mobile robot to autonomously navigate dynamic environments with real-time obstacle avoidance, monitor its internal bin capacity, and intelligently perform location-aware unloading without any human intervention.
Jayasekara R.G.S. IT22292940	Sorting the classified garbage into the correct disposal bin using a stationary robotic arm and monitoring the disposal bin levels continuously to notify the staff via mobile app.	<u>Stationary Robotic Arm Garbage Classifier</u> 1) Build the Waste Item Sorting Arm: Design and assemble a robotic arm equipped with grippers or actuators to pick up waste items after classification. Use servo motors to control the movements of the arm. 2) Monitor the Garbage Disposal Bins: Install ultrasonic sensors on each bin to measure the fill level by detecting the distance between the sensor and the waste surface.	Compared to the existing system, this system is fully automating the waste sorting and monitoring process without relying on human intervention. Also, sensors monitor bin fill levels in real time, and when full, the system sends automatic alerts to staff via a mobile app.

		3) Notify Garbage Cleaners when Any Bin is Full: Develop a mobile app connected to a cloud database to display the live status of each bin and send real-time notifications for easy staff monitoring.	
Nuwanthi B.D.T. IT22566102	Train a machine learning model to automatically classify waste into Plastic, Paper, Glass, and Metal when the collecting robot unloads the garbage.	<u>Machine Learning Model for Waste Classification</u> 1) Image Dataset Preparation: Collect waste item images from railway environments and label them into predefined categories (Plastic, Paper, Glass, Metal). Apply preprocessing techniques such as resizing and noise removal to prepare the dataset for training. 2) Model Development and Evaluation Design and train a CNN-based image classification model using the prepared dataset. Evaluate the model's performance through validation and testing to ensure accuracy and reliability for real-world scenarios. 3) Model Optimization and Deployment Optimize the trained model into a lightweight version suitable for edge devices. Deploy the model at the disposal platform and integrate it with the system for real-time waste classification during sorting.	This component helps the system automatically identify and separate waste types without human help, making the waste management process smarter and more efficient.

9. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
Perera W.B.N. IT22566348	<u>Autonomous Garbage Detection and Collection</u> This component aligns strongly with the Information Technology (IT) specialization by applying computer vision, embedded systems, and automated actuation to solve a real-world problem with minimal human intervention. • Real-time garbage detection through custom-trained object detection models demonstrates the application of computer vision, embedded AI, and intelligent edge computing within IT. • Robotic arm activation based on detection events reflects core IT skills such as microcontroller programming and software-controlled actuation. • Autonomous identification and collection of waste without human input showcase IT principles of intelligent automation and decentralized decision-making. • The seamless interaction between vision, logic, and actuation highlights integration of software logic with hardware interfaces, a key area in IT system design. Overall, this component exemplifies the IT specialization through its implementation of vision-based automation, embedded control logic, and intelligent interaction with real-world environments.
Sendanayaka H.K. IT22582324	<u>Autonomous Navigation and Intelligent Unloading Process</u> This component demonstrates deep integration of Information Technology (IT) concepts through real-time navigation, environment sensing, and intelligent task automation. • Obstacle-avoiding navigation uses real-time sensor data processing and control logic, emphasizing embedded systems and adaptive automation in IT. • Internal bin level monitoring and automatic unloading reflect condition-based decision-making and event-driven programming, fundamental IT concepts.

	• Disposal area identification via GPS modules illustrates geolocation-based logic and IoT implementation, relevant to smart, location-aware IT solutions. • Autonomous execution of the entire unload-resume cycle showcases embedded intelligence and real-world task automation, core to IT specialization. This component effectively demonstrates Information Technology specialization's application in smart automation, decentralized control, and IoT-based real-time decision-making in autonomous systems.
Jayasekara R.G.S. IT22292940	<u>Stationary Robotic Arm Garbage Classifier</u> This robotic sorting component aligns directly with the Information Technology (IT) specialization by integrating machine learning, embedded computer vision, autonomous IoT-based control, and intelligent decision-making - all core areas of modern IT systems. • Real-time image-based classification using a camera and a locally deployed machine learning model demonstrates the application of embedded AI, and intelligent visual processing key areas within the IT specialization. • The robotic arm responds to classification output and dynamically sorts waste into categorized bins (e.g., Plastic, Paper, Glass, Metal), showcasing real-time system control, automation logic, and embedded actuator programming - essential IT competencies in microcontroller-based automation. • The coordination between classification results, actuator control, and bin mapping reflects real-time data processing, event-driven logic, and embedded system communication - central pillars of IT system implementation and hardware software interfacing. Overall, this component represents a practical IT solution that combines edge-based machine learning inference, intelligent vision sensing, and adaptive embedded automation. It embodies its emphasis on building intelligent, reactive systems capable of operating autonomously in dynamic real-world environments in this case, supporting cleaner and smarter railway station waste management infrastructure.

Nuwanthi B.D.T.

IT22566102

Machine Learning Model for Waste Classification

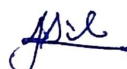
In this component, a machine learning model is developed to automatically classify collected waste into categories such as Plastic, Paper, Glass, and Metal. This directly aligns with Information Technology specialization, particularly in machine learning, artificial intelligence, and data processing.

The model is trained using waste images collected from railway stations. With the help of Convolutional Neural Networks (CNNs), the system learns to recognize different waste types. After improving the dataset through preprocessing and augmentation, the model is tested and optimized for real-world use. Finally, it is converted into a lightweight format (e.g. TensorFlow Lite) for smooth performance.

- Develop and train a CNN model to classify waste with high accuracy.
- Optimize the model for edge deployment to ensure efficient, real-time operation.
- Integrate the model into the disposal station for automatic waste sorting after collection.

This component supports a smart, automated waste management system and demonstrates strong IT skills in AI, embedded systems, and real-time integration.

10. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor	Mrs	Shashika	Lokuliyana	
Co-Supervisor	Mrs.	Suranjini	Silva	
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

This part is to be filled by the Topic Screening Staff members.

- a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes	☒	No	☐
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- b) Does the proposed topic exhibit novelty?

Yes	☒	No	☐
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- c) Do you believe they have the capability to successfully execute the proposed project?

Yes	☒	No	☐
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- d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes	☒	No	☐
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- e) Supervisor's Evaluation and Recommendation for the Research topic:

Recommended.

Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	✓
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

D Rewrite the objectives ~~and~~, tasks & novelty component distributing the scope among the members.

Staff Member's Name	Signature
Shashika Lokuliyana	

***Important:**

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.